

MAHMOUD OSAMA MOHAMED

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 2+ years of hands-on experience building responsive, high-performance web applications and scalable APIs. Skilled in React.js, Next.js, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, REST APIs, GraphQL, Socket.io, and modern UI frameworks. Experienced in translating Figma designs into production-ready interfaces, optimizing performance and load times, implementing secure JWT authentication, and collaborating in Agile teams to deliver maintainable full-stack solutions.

TECHNICAL SKILLS

Frontend: JavaScript ES6+, TypeScript, React.js, Next.js, React 19, Redux Toolkit, Zustand, Context API, TanStack Query, TanStack Router, Tailwind CSS, Bootstrap, Shadcn UI, Material UI, HTML5, CSS3

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL APIs, Socket.io, JWT Authentication, CRUD Operations, API Integration

Databases & ORMs: PostgreSQL, MongoDB, Prisma, Mongoose, Redis, Firebase Firestore, Supabase

Architecture & Concepts: Object-Oriented Programming, SOLID Principles, Clean Architecture, N-Tier Architecture, Onion Architecture, Repository Pattern, Unit of Work, Design Patterns

Tools & DevOps: Docker, Git, GitHub, GitHub CI/CD, VS Code, npm, Vite, Vercel, Netlify, Apache ECharts, Monaco Editor

Soft Skills: Agile Collaboration, Problem Solving, Backend Optimization, Task Ownership, Time Management

PROFESSIONAL EXPERIENCE

Full Stack Developer | Digital Pioneers Initiative (Digilians) - Egyptian Military Academy

Cairo, Egypt | Jan 2026 - Present

- Architected end-to-end full-stack integrations and structural database models, scaling system architectures to smoothly handle a 20% increase in concurrent client requests without downtime.
- Engineered scalable server-side infrastructures using Node.js and Express.js, optimizing database indexing to reduce API latency by 25% and ensure high-throughput data interchange.
- Optimized data-interchange processes and query schemas across relational/NoSQL databases, successfully slashing database query execution times by 25%.
- Collaborated within cross-functional Agile teams to implement asset optimization workflows, achieving a 15% reduction in production bundle sizes and significantly improving end-to-end API latency.

Front-End Developer Trainee | Information Technology Institute (ITI)

Benha, Egypt | Aug 2025 - Sep 2025

- Streamlined React.js/Next.js architectures and component libraries, increasing UI rendering speeds by 30% through modular atomic structures and advanced state synchronization patterns.
- Improved rendering paths and component performance, improving client-side responsiveness and user experience metrics by 30%.
- Achieved 1st place among graduation training projects by engineering and presenting StreamFlix, a full-featured media streaming application.

PROJECTS

Prep You.ai - AI-Powered Technical Interview Preparation Hub | [Live Demo](#)

Tech Stack: Next.js 15, React 19, TypeScript, Tailwind CSS, Zustand, Apache ECharts, GraphQL, REST APIs, Monaco Editor

- Developed an AI-driven behavioral and technical interview simulator with a responsive dashboard, code-execution sandboxes, and live feedback workflows.
- Built an audio-first behavioral interview component using animated SVG wireframe particle spheres to represent real-time voice frequencies.
- Integrated a Monaco-based code sandbox with custom dark themes to remove white flashes and improve the coding experience.
- Visualized scoring insights with Apache ECharts rings, radial gradients, and radar charts to compare user performance against community metrics.

ShopFlow E-Commerce | [GitHub Repo](#)

Tech Stack: React.js, Redux Toolkit, Tailwind CSS, Shadcn UI, Axios, Formik, Yup, JWT

- Programmed a full-featured e-commerce platform with secure JWT authentication, dynamic product filtering, form validation, and persistent cart features.
- Applied reusable atomic component structures to improve maintainability and reduce initial route load times by 20%.

StreamFlix Clone | [Live Demo](#)

Tech Stack: React.js, Tailwind CSS, Context API, REST APIs, Sonner, Supabase

- Constructed an interactive media streaming interface with modular React components and custom hooks, improving page performance by 30%.
- Integrated Supabase backend handlers for real-time catalog updates and dynamic client-side responses.

Personal Portfolio Development | [Live Demo](#)

Tech Stack: React.js, TypeScript, Tailwind CSS

- Designed and engineered a responsive personal portfolio application using React.js, TypeScript, and Tailwind CSS.
- Implemented a clean design system with custom Tailwind CSS theme tokens to improve visual consistency and maintainability.

EDUCATION

Bachelor's Degree in Mathematics and Computer Science | Faculty of Science

Benha University, Egypt | Sep 2021 - Jun 2025

- Built a strong foundation in programming, database concepts, problem solving, analytical thinking, and applied computer science.

CERTIFICATIONS

- MERN Stack Graduate Certificate | Digital Pioneers Initiative (Digilians) | 2026
- Full-Stack Software Engineering Certificate | Egyptian Military Academy | 2026
- Meta Front-End Developer Certificate | Coursera | 2026
- Front-End Developer Summer Internship | Information Technology Institute (ITI) | 2024

LANGUAGES

Arabic: Native | English: Advanced